



Name: _____ Cohort/Batch: _____ Date: _____ Score: _____

CELERY PRACTICE SHEET - 10 CONCEPTUAL Q&A

10 Q&A on brokers, task routing, retries, Beat, and testing

TOTAL: 10 QUESTIONS

Q1 What is Celery and what role does a message broker play?

Threat Model

Q6 How do you route tasks to specific queues in Celery?

Observability

Q2 What is the difference between `delay()` and `apply_async()`?

Architecture

Q7 How does Celery handle task result storage?

Lifecycle

Q3 How does Celery handle task retries and what is `autoretry_for`?

Configuration

Q8 What is the Celery Flower monitoring dashboard?

Compliance

Q4 What is Celery Beat and how do you schedule a periodic task?

Hardening

Q9 How do you test Celery tasks without a running broker?

Testing

Q5 What are Celery chains, groups, and chords?

SSO / IdP

Q10 What are Celery soft and hard time limits?

Tuning



ANSWER KEY & EXPLANATORY SOLUTIONS

Comprehensive verified answers, best-practice configs, security controls & reference

VERIFIED SOLUTIONS

A1 Celery is a distributed task queue.

Mitigation

Celery is a distributed task queue. When you call `task.delay()`, Celery serializes the call and sends it to a broker (Redis or RabbitMQ). Worker processes consume messages from the broker, execute tasks, and optionally store results in a backend.

A6 Define queues in the `task_queues` config.

Observability

Define queues in the `task_queues` config. Set `task_routes = {'myapp.tasks.heavy': {'queue': 'heavy_queue'}}`. Start workers with `--queues heavy_queue` to consume from that queue. This segregates slow tasks from fast ones for independent scaling.

A2 `task.delay(arg1, arg2)` is shorthand for

Primitives

`task.delay(arg1, arg2)` is shorthand for `apply_async((arg1, arg2))`. `apply_async` gives you full control: `countdown` (delay in seconds), `eta` (absolute time), `queue` name, `priority`, `retry` policy, and `task id` override.

A7 Set `result_backend = 'redis://localhost'` (or

Automation

`Set result_backend = 'redis://localhost' (or 'db+postgresql://...'). After completion, AsyncResult(task_id).get() blocks until the result is available. Results are stored for result_expires seconds (default 24 h) then purged.`

A3 Call `self.retry(exc=e, countdown=5,`

Hardening

Call `self.retry(exc=e, countdown=5, max_retries=3)` inside a task on failure. `@app.task(autoretry_for=(RequestException,), retry_backoff=True, max_retries=5)` automates this: Celery retries on the listed exceptions with exponential backoff.

A8 Flower is a real-time web UI

Governance

Flower is a real-time web UI for Celery. Run `celery -A app flower`. It displays active workers, task rates, queue lengths, failed task details, and lets you revoke pending tasks or rate-limit workers from the browser.

A4 Celery Beat is a scheduler that

Gotchas

Celery Beat is a scheduler that enqueues tasks on a schedule. Define entries in `beat_schedule`: `{'run-daily': {'task': 'app.tasks.cleanup', 'schedule': crontab(hour=0, minute=0)}}`. Beat requires a persistent store (db or Redis) to track last-run times.

A9 Set `task_always_eager = True` in the

Assessment

Set `task_always_eager = True` in the test config. This causes `.delay()` and `.apply_async()` to execute the task synchronously in the calling process without a broker. For unit tests of task logic only, call the function directly: `mytask(args)` skipping Celery entirely.

A5 A `chain(t1.s(), t2.s())` pipes output of

Federation

A `chain(t1.s(), t2.s())` pipes output of `t1` as input to `t2`. A `group([t1.s(), t2.s()])` runs tasks in parallel. A `chord(group, callback)` runs the group in parallel then passes all results to `callback` useful for map-reduce-style workflows.

A10 `soft_time_limit` raises `SoftTimeLimitExceeded`

Optimization

`soft_time_limit` raises `SoftTimeLimitExceeded` inside the task, giving it a chance to clean up. `hard_time_limit` (`task_time_limit`) sends `SIGKILL` to the worker process if the task exceeds the limit, with no cleanup possible. Prefer soft limits.